

Year 12

Mathematics Extension 1

HSC Trial Examination

2011

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Total Marks – 84

Attempt Questions 1 - 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1	(12 marks)	Marks
(a)	Factorise completely $x^4 + 2x^3 - 2x^2 - 4x$	2
(b)	Find the exact value of $\cos 15^\circ$.	2
(c)	The interval joining the points $A(9,12)$ and $B(1,b)$ is divided internally in the ratio 2:1 by the point $(a,0)$. Find a and b .	3
(d)	Solve $\frac{3}{x-2} \leq 4$	3
(e)	Find $\int \frac{dx}{\sqrt{36-x^2}}$	2

Question 2 (12 marks) Use a separate writing booklet **Marks**

- (a) If α , β and γ are the roots of $x^3 - 4x + 1 = 0$ find:
- | | | |
|-------|---|----------|
| (i) | $\alpha + \beta + \gamma$ | 1 |
| (ii) | $\alpha\beta\gamma$ | 1 |
| (iii) | $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ | 1 |
-
- (b) (i) Write $\sin x - \sqrt{3} \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and where $0 \leq \alpha \leq \frac{\pi}{2}$. **2**
- (ii) Hence, or otherwise solve the equation $\sin x - \sqrt{3} \cos x = 1$ for $0 \leq x \leq 2\pi$. **2**
-
- (c) $P(2at, at^2)$ is any point on the parabola $x^2 = 4ay$. The line k is parallel to the tangent at P and passes through the focus S of the parabola.
- | | | |
|-------|---|----------|
| (i) | Show that the equation of the line k is $y = tx + a$ | 2 |
| (ii) | The line k intersects the x -axis at the point Q . Find the coordinates of the midpoint, M , of the interval QS . | 2 |
| (iii) | Find the equation of the locus of M . | 1 |

Question 3 (12 Marks) Use a separate writing booklet **Marks**

- (a) Let $f(x) = e^{x+2}$
- (i) Explain why $f(x)$ has an inverse function. **1**
- (ii) Find the inverse function $f^{-1}(x)$ **1**
- (iii) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same number plane. Show all the important features. **2**
- (b) (i) Show that $e^x - 2x^2 = 0$ has a root in the interval $2 < x < 3$. **2**
- (ii) One approximate solution of the equation $e^x - 2x^2 = 0$ is $x = 2.5$. Use one application of Newton's method to find another approximation to this solution. Give your answer correct to three decimal places. **2**
- (c) (i) Prove that $\frac{\sec^2 x}{\tan x} = \frac{\operatorname{cosec} x}{\cos x}$ **2**
- (ii) Use the substitution $u = \tan x$ to evaluate in exact form **2**
- $$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec x}{\cos x} dx$$

Question 4 (12 Marks) Use a separate writing booklet	Marks
(a) The polynomial $P(x) = x^3 + ax + b$ has $(x - 5)$ as a factor and has a remainder of -60 when divided by $(x + 5)$. Find the values of a and b .	3
(b) Use the method of mathematical induction to prove that $4^n + 8$ is divisible by 6 for $n \geq 1$.	3
(c) (i) Sketch the graph of $y = 3 \sin^{-1} \frac{x}{2}$, stating its domain and range.	3
(ii) Show that the area of the region bounded by $y = 3 \sin^{-1} \frac{x}{2}$, the x axis and the line $x = 1$ is given by $A = \frac{\pi}{2} + 3\sqrt{3} - 6 \text{ units}^2$.	3

Question 5 (12 Marks) Use a separate writing booklet**Marks**

- (a) Calculate the acute angle between the lines $x - 3y - 2 = 0$ and $x - 2y = 0$ to the nearest degree. **2**

- (b) A particle is moving in a straight line such that its position at a time t is given by the equation

$$x = 2 \cos\left(3t + \frac{\pi}{3}\right)$$

- (i) Show that it is undergoing simple harmonic motion. **2**
- (ii) State the period and amplitude of motion. **2**

- (c) A bottle of water has a temperature of 20°C and is placed in a refrigerator whose temperature is 2°C . The cooling rate of the bottle of water is proportional to the difference between the temperature of the refrigerator and the temperature T of the bottle of water. This is expressed by the equation:

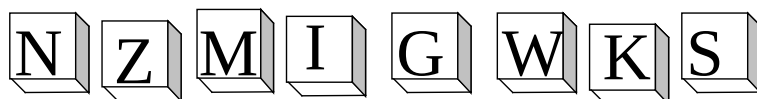
$$\frac{dT}{dt} = -k(T - 2)$$

where k is a constant of proportionality and t is the number of minutes after the bottle of water is placed in the refrigerator.

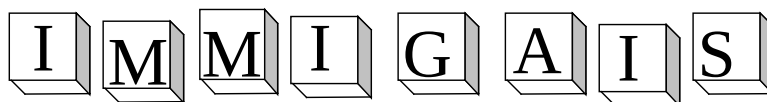
- (i) Show that $T = 2 + Ae^{-kt}$ satisfies the equation. **1**
- (ii) After 20 minutes in the refrigerator the temperature of the bottle of water is 10°C . What is the value of A and k in the above equation? **3**
- (iii) How long will it take for the bottle of water to cool down to 5°C ? Give your answer to the nearest minute. **2**

Question 6 (12 Marks) Use a separate writing booklet**Marks**

- (a) In a word game, Carlo draws the 8 letter tiles shown below which he must arrange to make "words".



- | | | |
|-------|--|----------|
| (i) | In how many distinct ways can Carlo arrange the eight tiles in a line to make eight-letter "words"? | 1 |
| (ii) | Carlo randomly chooses four tiles from the eight. How many distinct ways can Carlo arrange the four tiles in a line to make four-letter "words"? | 1 |
| (iii) | Carlo randomly chooses four tiles from his 8 to swap with a second player. How many different groups of four tiles (in no particular order) can Carlo choose? | 1 |
| (iv) | After swapping with the second player Carlo has the eight tiles below. In how many distinct ways can Carlo arrange these eight tiles in a line to make eight-letter "words"? | 1 |

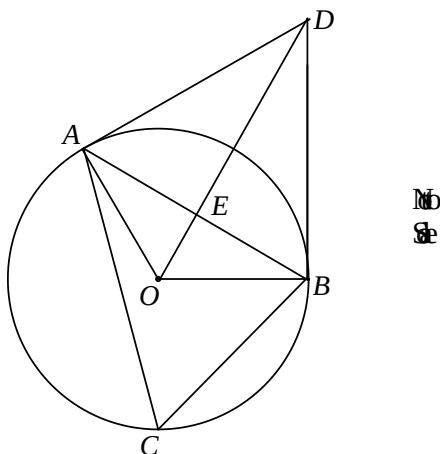


- | | | |
|-----|--|----------|
| (v) | In how many ways can the 8 tiles be arranged so that the "word" formed would start and end with M? | 1 |
|-----|--|----------|
- (b) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 2x \, dx$ as an exact value. **2**

Question 6 continues on the next page.

Question 6 continued.**Marks**

- (c) The diagram shows points A , B and C on a circle centre O . Tangents are drawn from A and B which meet at D . O is joined to D and the interval OD intersects AB at E .



Copy or trace the diagram into your workbook.

- | | | |
|-------|---|----------|
| (i) | Prove that $\angle AOB = 2 \times \angle DAB$. | 2 |
| (ii) | Prove that $AOBD$ is a cyclic quadrilateral. | 1 |
| (iii) | Prove that E is the midpoint of AB . | 2 |

Question 7 (12 Marks) Use a separate writing booklet**Marks**

- (a) A ball is thrown from the origin O with a velocity V and angle of elevation of θ , where $\theta \neq \frac{\pi}{2}$. You may assume that

$$x = Vt \cos \theta \text{ and } y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

where x and y are the horizontal and vertical displacements of the ball in metres from the O at time t seconds after being thrown.

- (i) Let $h = \frac{V^2}{2g}$ and show the equation of flight of the ball is **3**

$$y = x \tan \theta - \frac{1}{4h}x^2(1 + \tan^2 \theta)$$

- (ii) The point of intersection when two balls are thrown with an angle of elevation of θ_1 and θ_2 is (a, b) . **3**

Show that $a^2 < 4h(h - b)$.

- (b) (i) Find all the solutions to the inequality $\frac{x}{1-x^2} \geq 0$ **3**

- (ii) Show that $\tan x \sec x = \frac{\sin x}{1 - \sin^2 x}$ **1**

- (iii) Find all the solutions to $\tan x \sec x \geq 0$ when $0 \leq x \leq 2\pi$ **2**

END OF EXAMINATION

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

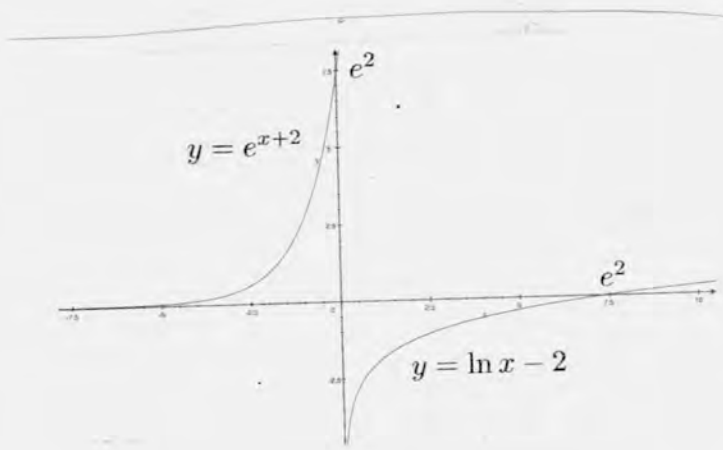
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

HSC Mathematics Extension 1 Yearly Examination

	Solution	Criteria
1(a)	$x^4 + 2x^3 - 2x^2 - 4x = x^3(x+2) - 2x(x+2)$ $= (x^3 - 2x)(x+2)$ $= x(x^2 - 2)(x+2)$	2 Marks: Correct answer. 1 Mark: Finds one factor.
1(b)	$\cos(45 - 30) = \cos 45 \cos 30 + \sin 45 \sin 30$ $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$ $= \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} + \frac{1}{2} \right)$ $= \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4}$	2 Marks: Correct answer. 1 Mark: Uses difference formula or states one correct exact ratio.
1(c)	$x = \frac{mx_2 + nx_1}{m+n} \qquad y = \frac{my_2 + ny_1}{m+n}$ $a = \frac{2 \times 1 + 1 \times 9}{2+1} \qquad 0 = \frac{2 \times b + 1 \times 12}{2+1}$ $= \frac{11}{3} \qquad 0 = 2b + 12$ $\qquad \qquad \qquad b = -6$	3 Marks: Correct answer. 2 Marks: Correctly finds either a or b . 1 mark: Uses ratio division formula.
1(d)	$(x-2)^2 \times \frac{3}{(x-2)} \leq 4 \times (x-2)^2$ $(x-2)3 \leq 4(x-2)^2 \quad x \neq 2$ $(x-2)(3-4x+8) \leq 0$ $(x-2)(11-4x) \leq 0$ $x < 2 \text{ and } x \geq \frac{11}{4}$	3 Marks: Correct answer. 2 Marks: Finds one correct solution. 1 Mark: Multiplies both sides of the inequality by $(x-2)^2$.
1(e)	$\int \frac{dx}{\sqrt{36-x^2}} = \sin^{-1} \frac{x}{6} + C$	2 Marks: Correct answer. 1 Mark: Recognises inverse sine function.

	QUESTION 2	
2(a) (i)	$x^3 + 0x^2 - 4x + 1 = 0$ $\alpha + \beta + \gamma = -\frac{b}{a} = -\frac{0}{1} = 0$	1 Mark: Correct answer.
2(a) (ii)	$\alpha\beta\gamma = -\frac{d}{a} = -\frac{1}{1} = -1$	1 Mark: Correct answer.
2(a) (iii)	$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{-4}{1} = -4$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$ $= \frac{-4}{-1}$ $= 4$	1 Mark: Correct answer.
b)(i)	If $\sin x - \sqrt{3} \cos x = R \sin(x - \alpha)$ then $R = \sqrt{1^2 + (\sqrt{3})^2} \text{ and } \tan \alpha = \sqrt{3}$ $\therefore R = 2 \quad \text{and} \quad \alpha = \frac{\pi}{3}$ So $\sin x - \sqrt{3} \cos x = 2 \sin\left(x - \frac{\pi}{3}\right)$	1 mark for R and 1 mark for α
b) (ii)	$\sin x - \sqrt{3} \cos x = 1$ $2 \sin\left(x - \frac{\pi}{3}\right) = 1$ $\sin\left(x - \frac{\pi}{3}\right) = \frac{1}{2}$ $x - \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}$ $x = \frac{\pi}{2}, \frac{7\pi}{6}$	1 for writing in form $\sin\left(x - \frac{\pi}{3}\right) = \frac{1}{2}$ 1 for solving for x

	QUESTION 3		
3(a)(i)	$f'(x) = e^{x+2} > 0$ for all x , ie $f(x)$ is monotonic increasing for all x and hence an inverse function exists Or for every value of y there is only one value of x , ie there is a 1-1 correspondance between y and x		1 mark for adequate explanation. Note that 'passes the horizontal line test' is not adequate without further explanation of what this means.
3(a)(ii)	$f(x) = e^{x+2}$ or $y = e^{x+2}$ Inverse function is $x = e^{y+2}$ $\log_e x = y + 2$ $y = \log_e x - 2$ $f^{-1}(x) = \log_e x - 2$		1 Mark: Correct answer.
3(a)(ii)			2 Marks: Correct answer. 1 Mark: Draws one of the graphs correctly.
3(b)(i)	$f(x) = e^x - 2x^2$ $f(2) = e^2 - 2(2)^2 = -0.6109... < 0$ $f(3) = e^3 - 2(3)^2 = 2.0855... > 0$ Since sign change from $x=2$ to $x=3$ and $f(x)$ is continuous in this interval (since $x > 0$) then a root exists $2 < x < 3$.	2	1 mark for showing change of sign 1 mark for statement of test. Must include continuity and why continuous.

3(b)(ii)	Root of $y = e^x - 2x^2$ using $x = 2.5$ $f(x) = e^x - 2x^2$ $f'(x) = e^x - 4x$ $f(2.5) = -0.318$ $f'(2.5) = 2.182$ $x_1 = 2.5 - \frac{f(2.5)}{f'(2.5)}$ $= 2.5 - \frac{-0.318}{2.182}$ $= 2.5 - (-0.145)$ $= 2.645$	2	1 for values of $f(2.5)$ and $f'(2.5)$ 1 use of N M to obtain answer.
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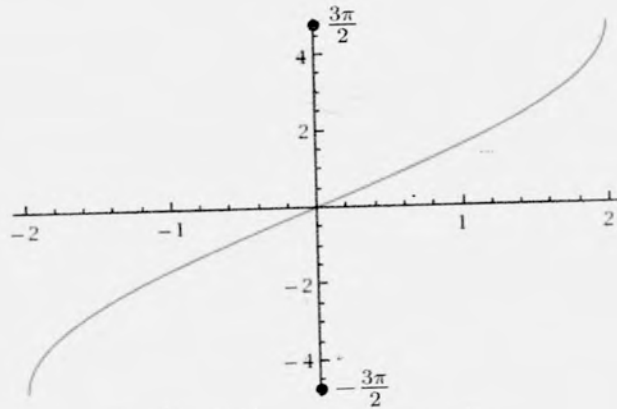
3(c)(i)	$\text{LHS} = \frac{\sec^2 x}{\tan x}$ $= \frac{1}{\cos^2 x} \div \tan x$ $= \frac{1}{\cos^2 x} \times \frac{\cos x}{\sin x}$ $= \frac{1}{\cos x} \times \operatorname{cosec} x$ $= \frac{\operatorname{cosec} x}{\cos x}$ $= \text{RHS}$	2 Marks: Correct answer. 1 Mark: Makes significant progress towards the solution.
3(c)(ii)	$u = \tan x \qquad u = \tan \frac{\pi}{3} = \sqrt{3} \qquad u = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$ $du = \sec^2 x dx$ $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\operatorname{cosec} x}{\cos x} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan x} dx$ $= \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{u} du$ $= [\log_e u]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}$ $= \log_e \sqrt{3} - \log_e \frac{1}{\sqrt{3}}$ $= \log_e 3$	2 Marks: Correct answer. 1 Mark: Recognises the use of part (i) or makes progress in the substitution.

QUESTION 4			
4(a)	$p(x) = x^3 + ax + b$ has $(x - 5)$ as one of its factors $\therefore p(5) = 0$ $125 + 5a + b = 0$ Remainder = -60 when divided by $(x + 5)$. $\therefore p(-5) = -60$ $-125 - 5a + b = -60$ Solve simultaneously to find a and b. $5a + b + 125 = 0$ ① $-5a + b - 65 = 0$ ② $2b + 60 = 0$ ① + ② $2b = -60$ $b = -30$ $5a - 30 + 125 = 0$ $5a = -95$ $a = -19$	3	1 mark for use of factor theorem 1 mark for use of remainder theorem 1 mark for solving simultaneously Part marks as appropriate if other approaches taken

b)	Step 1 show true of $n = 1$ $4^1 + 8 = 12$ 12 is divisible by 6 $\therefore$ true for $n = 1$ Step 2 Assume true for $n = k$ Ie $4^k + 8 = 6p$ (p is a positive integer) Step 3: hence prove true for $n = k + 1$ $4^{k+1} = 6q \text{ (} q \text{ is a positive integer)}$ $\begin{aligned} LHS &= 4^{k+1} + 8 \\ &= 4(4^k) + 8 \\ &= 4(4^k + 8 - 8) + 8 \\ &= 4(4^k + 8) - 32 + 8 \\ &= 4(6p) - 24 \\ &= 24p - 24 \\ &= 6(4p - 4) \\ &= 6q \text{ (} q \text{ is positive integer } > 1 \text{ since } p > 1) \\ &= RHS \end{aligned}$ Conclusion Using the principle of induction Since true for $n = 1$, and since if true for $n = k$ is also true for $n = k + 1$, by induction is true for all $n \geq 1$.	4	1 mark for step 1 2 marks for step 3 Or 1 mark for significant progress or a simple error in step 3
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(c)(i)

$$y = 3 \sin^{-1} \frac{x}{2}$$

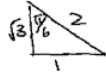
Domain: $-2 \leq x \leq 2$ Range: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ 1 mark each
domain and
range and
correct graph

c)(ii)

Area under $y = 3 \sin^{-1} \frac{x}{2}$ from $x=0$ to $x=1$.

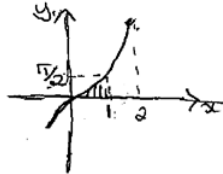
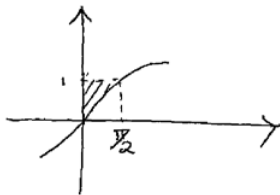
$$x=0 \quad y=0$$

$$x=1 \quad y = 3 \sin^{-1} \left(\frac{1}{2} \right)$$



$$= 3 \cdot \frac{\pi}{6}$$

$$= \frac{\pi}{2}$$

Inverse of $y = 3 \sin^{-1} \frac{x}{2}$ is $y = 2 \sin \frac{x}{3}$.

Area needed is the same as rectangle

$$1 \times \frac{\pi}{2} - \int_0^{\pi/2} 2 \sin \frac{x}{3} dx.$$

$$\therefore A = \frac{\pi}{2} - 2 \int_0^{\pi/2} \sin \frac{x}{3} dx$$

$$= \frac{\pi}{2} + 6 \left[\cos \frac{x}{3} \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} + 6 \left[\cos \frac{\pi}{6} - \cos 0 \right]$$

$$= \frac{\pi}{2} + 6 \cdot \frac{\sqrt{3}}{2} - 6$$

$$= \frac{\pi}{2} + 3\sqrt{3} - 6.$$

2 marks
or 1 mark for
evidence of working
towards correct
solution

1 mark.

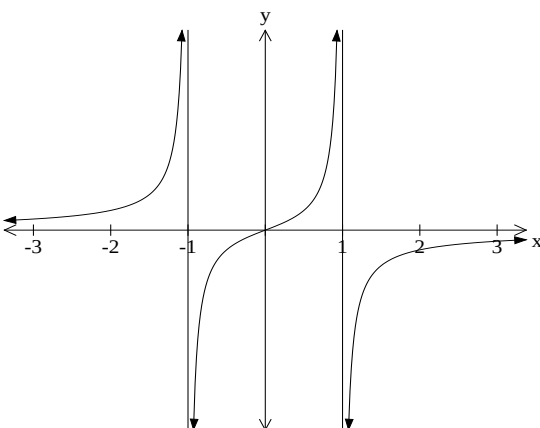
QUESTION 5			
5(a)	$x - 3y - 2 = 0$ $3y = x - 2$ $y = \frac{x}{3} - \frac{2}{3}$ $x - 2y = 0$ $2y = x$ $y = \frac{x}{2}$ $\text{Gradient} = \frac{1}{3}$ $\text{Gradient} = \frac{1}{2}$. $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $= \left \frac{\frac{1}{2} - \frac{1}{3}}{1 + \frac{1}{2} \cdot \frac{1}{3}} \right $ $= \left \frac{\frac{1}{6}}{\frac{7}{6}} \right $ $= \frac{1}{7}$ $\theta = \tan^{-1} \left(\frac{1}{7} \right)$ $= 8^\circ$ (nearest degree)	2	1 for gradients of the two lines 1 for evaluating the angle
b) i)	$x = 2 \cos \left(3t + \frac{\pi}{3} \right)$ $\dot{x} = -2 \sin \left(3t + \frac{\pi}{3} \right) \cdot 3$ $\dot{x} = -6 \sin \left(3t + \frac{\pi}{3} \right)$ $\ddot{x} = -6 \cos \left(3t + \frac{\pi}{3} \right) \cdot 3$ $= -18 \cos \left(3t + \frac{\pi}{3} \right)$ $\ddot{x} = -9 \left(2 \cos \left(3t + \frac{\pi}{3} \right) \right)$ $\ddot{x} = -3^2 x$ which is of the form $\ddot{x} = -n^2 x$ $\therefore$ particle is in SHM	2	1 mark for correct differentiations 1 mark for writing in the correct form and stating a conclusion
ii)	Period $= \frac{2\pi}{n}$ $= \frac{2\pi}{3}$ Amplitude = 2 units	1 1	

5(c) (i)	$T = 2 + Ae^{-kt} \quad \text{or} \quad Ae^{-kt} = T - 2$ $\frac{dT}{dt} = -kAe^{-kt}$ $= -k(T - 2)$	1 Mark: Correct answer.
5(b) (ii)	Initially $t = 0$ and $T = 20$ $T = 2 + Ae^{-kt}$ $20 = 2 + Ae^{-k \times 0}$ $A = 18$ Also $t = 20$ and $T = 10$ $T = 2 + 18e^{-kt}$ $10 = 2 + 18e^{-k \times 20}$ $e^{-k \times 20} = \frac{8}{18}$ $-20k = \log_e \frac{4}{9}$ $k = -\frac{1}{20} \log_e \frac{4}{9}$ $= \frac{1}{20} \log_e \frac{9}{4}$ $= 0.04054651081\dots$	3 Marks: Correct answer. 2 Marks: Finds the value of A and an expression for k. 1 Mark: Finds the value of A.
5(b) (iii)	We need to find t when $T = 5$ $T = 2 + 18e^{-kt}$ $5 = 2 + 18e^{-kt}$ $e^{-kt} = \frac{3}{18}$ $-kt = \log_e \frac{1}{6}$ $t = \frac{1}{k} \log_e 6$ $= 20 \frac{\log_e 6}{\log_e \frac{9}{4}}$ $= 44.19022583$ It will take about 44 minutes for the bottle to cool to 5°C?	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.

[illegible]

(ii)	$\angle DAO = \angle DBO = 90^\circ$ (tangent perpendicular to radius) $\therefore \angle DAO + \angle DBO = 180^\circ$ (sum of two right angles) $\therefore$ opposite angles of $AOBD$ are supplementary $\therefore AOBD$ is a cyclic quadrilateral.	1	1 mark as long as statement that opposite angles are supplementary, and why.
(iii)	Aim: Prove that E is the midpoint of AB . $AO = BO$ (equal radii) $AD = BD$ (tangents from an external point are equal in length.) $\therefore AOBD$ is a kite $\therefore OD$ bisects AB (symmetry of a kite) $\therefore E$ is midpoint of AB .	2	Can also be done by isosceles triangles 2 marks for complete proof. 1 mark if some relevant facts are stated or proof is incomplete

	QUESTION 7	
7(a) (i)	$x = Vt \cos \theta \quad (1)$ $y = -\frac{1}{2}gt^2 + Vt \sin \theta \quad (2)$ From eqn (1) $t = \frac{x}{V \cos \theta}$ sub into eqn (2) $y = -\frac{1}{2}g\left(\frac{x}{V \cos \theta}\right)^2 + V\left(\frac{x}{V \cos \theta}\right)\sin \theta$ $= -\frac{gx^2}{2V^2 \cos^2 \theta} + \frac{\sin \theta x}{\cos \theta}$ $= -\frac{gx^2 \sec^2 \theta}{2V^2} + \tan \theta x$ $= -\frac{2gx^2 \sec^2 \theta}{4V^2} + \tan \theta x$ $= -\frac{x^2 \sec^2 \theta}{4h} + \tan \theta x \quad \text{Using } h = \frac{v^2}{2g}$ $= x \tan \theta - \frac{1}{4h}x^2(1 + \tan^2 \theta)$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 1 Mark: Makes t the subject of eqn (1) or equivalent progress.
7(a) (ii)	Now (a, b) satisfies the equation $y = x \tan \theta - \frac{1}{4h}x^2(1 + \tan^2 \theta)$ $b = a \tan \theta - \frac{1}{4h}a^2(1 + \tan^2 \theta)$ $4hb = 4ha \tan \theta - a^2(1 + \tan^2 \theta)$ $(1 + \tan^2 \theta)a^2 - 4ha \tan \theta + 4hb = 0$ $a^2 \tan^2 \theta - 4ha \tan \theta + 4hb + a^2 = 0$ Quadratic equation has 2 solutions if the discriminate is greater than zero. $b^2 - 4ac > 0$ $(-4ha)^2 - 4a^2(4hb + a^2) > 0$ $16h^2a^2 - 16a^2hb - 4a^4 > 0$ $4a^2(4h^2 - 4hb - a^2) > 0$ $4h^2 - 4hb - a^2 > 0$ $a^2 < 4h^2 - 4hb$ $a^2 < 4h(h - b)$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 1 Mark: Substitutes (a, b) into equation of flight and simplifies.

7(b) (i)	$(1-x^2)^2 \times \frac{x}{1-x^2} \geq 0 \times (1-x^2)^2 \quad x \neq \pm 1$ $(1-x^2)x \geq 0$ $(1-x)(1+x)x \geq 0$  From the graph $0 \leq x < 1$ and $x < -1$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 1 Mark: Finds part of the solution or demonstrates some understanding.
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7(c) (ii)	$\begin{aligned} \text{LHS} &= \tan x \sec x \\ &= \frac{\sin x}{\cos x} \times \frac{1}{\cos x} \\ &= \frac{\sin x}{\cos^2 x} \\ &= \frac{\sin x}{1 - \sin^2 x} \\ &= \text{RHS} \end{aligned}$	1 Mark: Correct answer.
7(c) (iii)	Using parts (i) and (ii) $0 \leq \sin x < 1 \quad \text{and} \quad \sin x < -1$ $0 \leq x < \frac{\pi}{2} \quad \text{No solution}$ Solution also in the second quadrant $0 \leq x < \frac{\pi}{2} \text{ and } \frac{\pi}{2} < x \leq \pi$	2 Marks: Correct answer. 1 mark: Makes some progress using parts (i) and (ii).